turgor

code + validation report

# See soybean drought losses weeks before harvest

A crop-physiology stress index for co-ops, input resellers, lenders and insurers in Brazil. Satellite and weather data only, no farm sensors.

## Problem

Soybean losses below trend average **R$ 700–1,200 per ha a year** in Paraná and Rio Grande do Sul (2016–2025, at today's price). Those who finance the crop learn how bad a season was at harvest, through field inspections. Rainfall triggers for parametric insurance miss 44% of real losses; the official ZARC zoning barely separates loss seasons (AUC 0.57).

## Product

Weekly, December to April: each field's crop stage from Sentinel-2 and MODIS, plus water deficit, heat above 30 °C and their interaction, weighted by crop stage (flowering and seed fill count 3×). Output: a stress map per municipality or field, and expected loss in R$ per ha. Live monitor from 2 December 2026.

## Evidence (public, reproducible; 780 municipalities, RS + PR)

| | Turgor | Rainfall | NDVI | ZARC |
|---|---|---|---|---|
| Backtest 2001–2023, AUC | **0.86** | 0.78 | 0.77 | 0.57 |
| Backtest, share of losses missed | **0.33** | 0.44 | 0.44 | 0.64 |
| New seasons 2024–25, AUC | **0.67** | 0.56 | 0.61 | 0.57 |
| New seasons, share of losses missed | **0.36** | 0.51 | 0.42 | 0.46 |

**Early warning:** by 15 January the index matches what a rainfall index only knows at season end, 8–12 weeks before harvest. **Real season:** on 15 Feb 2025, municipalities rated "severe" ended 30% below trend; 83% lost more than 10%. **Limit:** in Rio Grande do Sul the municipal gain over rainfall is likely but not yet proven.

## Business

B2B, per hectare per season (hypotheses being tested: risk score R$ 2–6/ha; alerts R$ 1–3/ha; insurance trigger licence 1–3% of premium; portfolio backtests R$ 30–100 k). Market at R$ 3/ha: RS + PR R$ 37 M a year, Brazil R$ 146 M (48.7 M ha, Conab 2025/26).

## Looking for

**Pilots:** a free backtest on your own fields or portfolio, report in two weeks. **Cofounders:** a data and remote-sensing CTO; a business lead from rural credit or insurance. **Advisors:** crop modelling, agricultural insurance, rural credit.

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